

SE3082

AWS EC2 Lab - Screenshots and Calculations

Lab 06

Exercise 6: Parallelizing the element-wise multiplication of two arrays using OpenMP and the strip-mining technique

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On instance with 2 vCPUs (t3.micro)

Nano - creating the strip-mining code to perform element wise multiplication of two arrays

```
GNU nano 6.2                                ex6_stripmine.c
#include <stdio.h>
#include <stdlib.h>
#include <omp.h>

#define N 1000000
#define STRIP_SIZE 8

int main() {
    double *A = malloc(N * sizeof(double));
    double *B = malloc(N * sizeof(double));
    double *C = malloc(N * sizeof(double));
    double tstart, tstop, tcalc;

    for (int i = 0; i < N; i++) {
        A[i] = i * 1.0;
        B[i] = i * 2.0;
    }

    tstart = omp_get_wtime();

    #pragma omp parallel for
    for (int strip = 0; strip < N; strip += STRIP_SIZE) {
        int end = strip + STRIP_SIZE;
        if (end > N) end = N;
        #pragma omp simd
        for (int i = strip; i < end; i++) {
            C[i] = A[i] * B[i];
        }
    }

    tstop = omp_get_wtime();
    tcalc = tstop - tstart;

    printf("Threads used: %d\n", omp_get_max_threads());
    printf("C[0] = %f, C[500000] = %f, C[999999] = %f\n", C[0], C[500000], C[999999]);
    printf("Time = %f seconds\n", tcalc);

    free(A);
    free(B);
    free(C);
    return 0;
}
```

Output of running the strip-mining code to perform element wise multiplication of two arrays

```
[ec2-user@ip-172-31-17-201 ~]$ ./ex6_stripmine.o  
Threads used: 2  
C[0] = 0.000000, C[500000] = 5000000000000.000000, C[999999] = 19999960000002.000000  
Time = 0.005471 seconds
```